Cohort Batch Correction in SSBMF: Literature Summary and Implementation Outline

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The Problem

Supervised Sparse Bayesian Matrix Factorization (SSBMF) jointly fits a low-rank genomics factor model and a Cox proportional hazards survival model. The genomics reconstruction is $Y = LF' + E$ and the survival linear predictor is $\eta = L\beta$. When trained on a single cohort — TCGA or CPTAC individually — prognostic factors are recovered and external C-indices reach 0.55–0.67.

When TCGA (RNA-seq) and CPTAC (proteomics) are pooled, the dominant axis of variation in Y is platform technology, not biology. This technical factor captures the first principal component, and the survival signal is too weak to compete: CAVI converges with $\hat{\beta} \approx 0$, producing a prognostic score of no clinical value. Per-platform z-score normalization reduces but does not eliminate the collapse. A principled solution requires modeling the platform offset *within* the CAVI iteration so the survival component is shielded from it throughout inference.

Literature Context

Bayesian linear mixed models. The linear mixed model separates signal from structured nuisance variation through a hierarchical prior on group-level deviations. In the frequentist tradition, restricted maximum likelihood (REML) estimates variance components (Chen and Dunson 2003). In Bayesian formulations, Gelman (2006) recommended half-Cauchy priors on variance parameters to avoid the pathologies of inverse-gamma priors at small scales. More flexible sparsity-inducing priors include spike-and-slab (George and McCulloch 1993), the horseshoe (Carvalho, Polson, and Scott 2010), and its regularized variant (Piironen and Vehtari 2017), which truncates the slab to improve numerical stability for weakly identified parameters. These priors are most beneficial when borrowing strength across many groups ($C \gg 1$ with small group sizes); with $C = 2$ and $n_c \geq 100$, a simple Normal prior is well-identified and requires no hyperprior.

EBNM and EBMF — the methodological foundation of SSBMF. Stephens (2017) introduced the Empirical Bayes Normal Means (EBNM) framework, which estimates the prior family from the marginal likelihood rather than fixing it in advance. Wang and Stephens (2021) extended this to Empirical Bayes Matrix Factorization (EBMF): $Y = LF' + E$ is fit by coordinate ascent variational inference in which each column update reduces to an EBNM problem, selecting the appropriate level of sparsity automatically. The `ebnm` R package (Willwerscheid, Carbonetto, and Stephens 2025) provides the computational backend. SSBMF is a supervised extension of EBMF that adds a Cox partial likelihood on $\eta = L\beta$. The cohort column extension proposed here fits naturally within this infrastructure: the cohort column uses a simple Normal prior rather than EBNM (conjugate to the Normal likelihood, so the posterior is closed-form), and the fixed indicator values have zero posterior variance, bypassing the L-update entirely.

Genomics batch-effect correction methods. ComBat (Johnson, Li, and Rabinovic 2007) corrects additive batch shifts gene-by-gene via empirical Bayes shrinkage and is the de facto standard for bulk RNA-seq preprocessing; ComBat-seq (Zhang, Parmigiani, and Johnson 2020) extends this to count data. SVA (Leek and Storey 2007) estimates latent confounders when batch labels are unknown; RUV (Risso et al. 2014) uses control genes to identify technical factors. PEER (Stegle et al. 2010) is the closest published architectural precedent for the proposed approach: it models gene expression as a sum of *known* covariate effects (fixed indicator loadings, estimated gene-level magnitudes) and *unknown* latent factors within a single Bayesian factor model — exactly the structure of a fixed cohort column in L paired with an estimated F row. All five methods operate as preprocessing steps decoupled from the survival inference objective.

Among multi-omics factor analysis methods, MOFA (Argelaguet et al. 2018) is architecturally closest to SSBMF: it learns a shared low-dimensional representation across omics layers via variational Bayes with ARD sparsity priors. MOFA+ (Argelaguet et al. 2020) extends this to multiple sample groups. Harmony (Korsunsky et al. 2019) and scVI (Lopez et al. 2018) perform batch integration in embedding space. DIABLO (Singh et al. 2019) is supervised but targets categorical phenotypes and cannot constrain cohort effects out of a continuous survival score.

Why existing methods are insufficient. Every method above either decouples batch correction from the survival objective (ComBat, SVA, RUV, PEER), targets an unsupervised objective (MOFA, MOFA+, Harmony, scVI), or cannot prevent cohort effects from entering a prognostic score (DIABLO). None jointly optimizes a survival objective while estimating platform offsets within the same iteration. PEER is the closest architectural precedent but has no survival component; MOFA+ is the closest variational Bayes ancestor but has no equivalent of $\beta_{\text{cohort}} \equiv 0$.

Proposed Solution

Model. The cohort indicator column is appended directly to the existing L matrix, keeping a single L and F throughout:

$$L_{\text{aug}} = [\; \underbrace{L}_{n\times K} \;|\; \underbrace{\mathbf{c}}_{n\times 1} \;], \qquad F_{\text{aug}} = [\; \underbrace{F}_{K\times p} \;|\; \underbrace{\mathbf{f}_c}_{1\times p} \;]$$

where $\mathbf{c} \in \{0,1\}^n$ is the fixed cohort indicator (e.g., 0 = TCGA, 1 = CPTAC) and $\mathbf{f}_c \in \mathbb{R}^p$ is the estimated platform offset across all genes. With $C = 2$ cohorts this requires one extra column; for C cohorts, $C-1$ indicator columns (corner-point parameterization against a reference cohort) or C columns (all cohorts, identifiable because Y is column-centred before fitting). The survival linear predictor $\eta = L\beta$ uses only the original K columns:

need a column for each cohort? or C levels in a single column?

$$\eta_i = \sum_{k=1}^{K} L_{ik}\,\beta_k, \qquad \beta_{\text{cohort}} \equiv 0$$

The cohort column enters the genomics likelihood but never reaches the Cox partial likelihood. A patient from a new cohort at test time needs no cohort label — the prognostic score is defined purely by their projection onto the K biological factors.

Implementation

The key insight is that $\mathbf{c}$ is a **fixed** column of L (not estimated) with a **different prior** on the corresponding row of F (Normal, not point-Normal or point-exponential). This requires three targeted changes to the existing modular pipeline and leaves all other update functions structurally unchanged.

1. L-update: skip the cohort column. `update_L_all` (`code/update_L.R:267`) loops over $k = 1, \dots, K$. The cohort column $\mathbf{c}$ is fixed with zero posterior variance, so the loop simply does not include it — no change to `update_L_k` or the EBNM call.

2. F-update: Normal prior for the cohort row. Each global row of F is updated via EBNM through `update_F_k` (`code/update_F.R:94`). For the cohort row, the Normal prior $f_{cj} \sim \mathcal{N}(0, \sigma^2)$ is conjugate to the Normal likelihood, giving a closed-form posterior without any EBNM optimization: $A_{cj} = \tau_j n_c + \sigma^{-2}$, $B_{cj} = \tau_j \sum_{i:\, c_i=1} R_{ij}$, $\bar{f}_{cj} = B_{cj}/A_{cj}$, and $E[f_{cj}^2] = A_{cj}^{-1} + \bar{f}_{cj}^2$, where $R_{ij} = Y_{ij} - \sum_k L_{ik}\bar{F}_{kj}$ is the global-factor residual. In R this is `A <- n_c * Tau + 1/sigma^2` and `EF_cohort <- (t(c_vec) %*% R) * Tau / A`, operating on all p genes simultaneously. When computing the partial residual R^{-k} inside `compute_R_k` (`code/update_L.R:59`) for global factor updates, the cohort contribution must be subtracted first: `Y_adj <- Y - c %*% EF_cohort` is passed as the working response, with no structural change to `compute_R_k`.

3. tau-update: add cohort variance to Var_Term. `compute_var_term` (`code/update_tau.R:60`) accumulates posterior variance across all estimated components. Because $\mathbf{c}$ is fixed, the only new variance is A_{cj}^{-1} for patients in the indicator-positive cohort, added as `Var_cohort_n <- c %*% matrix(1/A_cohort, 1, p)` before the precision MLE.

4. beta-update and ELBO: no change. `update_beta_all` (`code/fit_modular.R:436`) loops over $k = 1, \dots, K$ and returns a K-vector — the cohort column has no β entry. The ELBO gains one closed-form Normal KL term summed over genes: $\frac{1}{2}\sum_j (E[f_{cj}^2]/\sigma^2 + \log(A_{cj}\sigma^2) - 1)$.

5. Initialization. To prevent the platform PC from being absorbed by a global factor before the cohort row can claim it: (i) initialize $\bar{f}_{cj}^{(0)}$ as the differential gene mean between cohorts (`colMeans(Y[c==1,]) - colMeans(Y[c==0,])`), then (ii) initialize L and F from SVD of the cohort-mean-subtracted Y.

Session Plan

Session	Deliverable	Acceptance gate
A	`derivations/qF_cohort/` — closed-form derivation (LaTeX + PDF)	Advisor math review
B	`code/update_F_cohort.R` + 15 unit tests + demo	15/15 tests pass
C	Modifications to `compute_R_k`, `compute_var_term`, `compute_elbo.R`, `fit_modular.R` + integration tests	~218 tests pass; `cohort_id=NULL` reproduces current results to 10^{-6}
D	Synthetic validation ($n = 200$, $p = 500$, $K = 3$, $C = 2$, known ground truth)	$\hat{\beta}$ recovery $r > 0.9$; C-index $\geq$ baseline and $\geq$ z-score
E	Real TCGA + CPTAC benchmark + results report	$\hat{\beta}$ non-collapse (≥ 2 components > 0.1); external C-index ≥ 0.60 on ≥ 2 of 4 held-out cohorts

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Full bibliography and mathematical derivations in `cohort_lmm_review_plan.pdf`.